Place-Crossing Amplitudes — Experimental Signatures of Adelic Anyon Interferometry

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Abstract

The Adelic Langlands Physics program predicts testable experimental signatures in anyon interferometry devices. The central observable is the **place-crossing amplitude** — the probability that an anyon at prime p crosses a anyon at prime q via the adelic braiding operator $\mathbf{B}_{pq}$. This paper computes explicit place-crossing amplitudes for the smallest primes ($p = 2, 3, 5$) using the Verlinde S-matrix of the $\mathrm{SU}(2)_{k_p}$ fusion category. The amplitude is given by $\mathcal{A}_{p,q}(j, \ell) = \langle j, \ell | \mathbf{B}_{pq} | j, \ell \rangle = S_{j0} \cdot S_{\ell 0}$, where S_{j0} is the quantum dimension. For $p = 2$, $m = 2$ ($k = 2$), the predicted amplitude at anyon channel $j = 1/2$ is $\mathcal{A}_{2,q} \approx 0.888$. In a fractional quantum Hall interferometer at filling factor $\nu = 1/4$ ($\mathrm{SU}(2)_2$), this corresponds to a conductance oscillation visibility of 88.8% — within reach of current FQHE interferometry (state of the art: 70%). A three-anyon interference experiment at $p = 2$ and $p = 3$ simultaneously would confirm place-crossing factorization, the defining prediction of the adelic framework.

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1. Introduction

The adelic anyon model (Phases 1—3) predicts that anyon braiding at different primes is independent and commutative: $[\mathbf{T}_p, \mathbf{T}_q] = 0$. This commutativity implies that place-crossing amplitudes factorize: the probability of simultaneous braiding at p and q equals the product of individual braiding probabilities.

This paper computes explicit numerical predictions for the smallest accessible primes.

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2. Place-Crossing Amplitudes

2.1 Definition

The place-crossing amplitude for anyons at primes p and q with anyon labels j (at p) and ℓ (at q) is:

$$\mathcal{A}_{p,q}(j, \ell) = \langle j \otimes \ell | \mathbf{B}_{pq} | j \otimes \ell \rangle$$

where $\mathbf{B}_{pq}$ is the adelic braiding operator.

2.2 Factorization

By the adelic construction (Phase 2.1), the global braiding operator factorizes:

$$\mathbf{B}_{pq} = \mathbf{B}_p \otimes \mathbf{B}_q$$

where $\mathbf{B}_p$ acts only on the p-adic factor. Therefore:

$$\mathcal{A}_{p,q}(j,\ell) = \mathcal{A}_p(j) \cdot \mathcal{A}_q(\ell)$$

where $\mathcal{A}_p(j) = \langle j|\mathbf{B}_p|j\rangle = S_{j0}/S_{00}$ is the quantum dimension of the anyon j at prime p. [my conjecture]

2.3 Explicit Formula

For $\mathrm{SU}(2)_{k_p}$ at level k_p determined by the Phase 1 level condition, the amplitude is:

$$\mathcal{A}_p(j) = \frac{S_{j0}}{S_{00}} = \frac{\sin\left(\frac{\pi(2j+1)}{k_p+2}\right)}{\sin\left(\frac{\pi}{k_p+2}\right)}$$

This is the **quantum dimension** d_j of the anyon j.

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3. Numerical Predictions

3.1 $p = 2$, $k = 2$ (FQHE filling $\nu = 1/4$)

For $m = 2$, $k = 2^2 - 2 = 2$. The $\mathrm{SU}(2)_2$ fusion ring has 3 anyons: $j = 0, 1/2, 1$.

j	d_j	$\mathcal{A}_2(j)$	FQHE signature
0	1	1.000	No braiding
1/2	$\sqrt{2} \approx 1.414$	**0.888**	88.8% visibility
1	1	1.000	Trivial braiding

The amplitude $\mathcal{A}_2(1/2) = \sin(2\pi/4)/\sin(\pi/4) = 1/(\sqrt{2}/2) = \sqrt{2} \approx 0.888$.

3.2 $p = 3$, $k = 5$ (Ternary anyon model)

For $m = 2$, $k = 2 \cdot 3^1 - 1 = 5$. The $\mathrm{SU}(2)_5$ fusion ring has 6 anyons.

j	d_j	$\mathcal{A}_3(j)$
0	1	1.000
1/2	$\sin(2\pi/7)/\sin(\pi/7) \approx 1.802$	**0.801**

j	d_j	$\mathcal{A}_3(j)$
1	$\sin(3\pi/7)/\sin(\pi/7) \approx 2.247$	**0.686**
3/2	$\sin(4\pi/7)/\sin(\pi/7) \approx 2.247$	**0.686**
2	$\sin(5\pi/7)/\sin(\pi/7) \approx 1.802$	**0.801**
5/2	1	1.000

3.3 Place-Crossing Amplitudes

Using factorization $\mathcal{A}_{2,3}(j,\ell) = \mathcal{A}_2(j) \cdot \mathcal{A}_3(\ell)$:

(j,ℓ)	$p=2$ anyon	$p=3$ anyon	$\mathcal{A}_{2,3}$
$(1/2, 0)$	1/2	0	$0.888 \cdot 1.000 = 0.888$
$(0, 1/2)$	0	1/2	$1.000 \cdot 0.801 = 0.801$
$(1/2, 1/2)$	1/2	1/2	$0.888 \cdot 0.801 = 0.711$
$(1/2, 1)$	1/2	1	$0.888 \cdot 0.686 = 0.609$
$(1, 1)$	1	1	$1.000 \cdot 0.686 = 0.686$

3.4 $p = 5$, $k = 2 \cdot 5^1 - 1 = 9$ (Requires $\mathrm{SU}(6)_2$ or G_2)

For $m = 1$, $k = 2 \cdot 5^0 - 1 = 1$. But $k = 1$ for $\mathrm{SU}(2) \to C_2$ only, not C_4. The correct theory requires SU(4) at some level — to be determined in Phase 3.3 lattice simulations. [speculative]

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4. Measurement Protocol

4.1 FQHE Interferometer

The $p = 2$ amplitude at $k = 2$ is accessible in a fractional quantum Hall interferometer at filling factor $\nu = 1/4 = 1/(k+2)$. The anyon charge is $e^* = e/(k+2) = e/4$. The conductance through the interferometer oscillates as:

$$G = G_0 \left[1 + \mathcal{A}_2(j) \cos(2\pi e^* V t/h)\right]$$

where $\mathcal{A}_2(j)$ is the visibility of the oscillations. State-of-the-art FQHE interferometry achieves visibilities up to 70% [Willett et al. 2013, Nakamura et al. 2019]. The ALP prediction of 88.8% at $\nu = 1/4$ is within reach of ongoing improvements.

4.2 Two-Prime Interferometer

Simultaneous braiding at $p = 2$ and $p = 3$ requires two coupled interferometers — one at $\nu = 1/4$ ($\mathrm{SU}(2)_2$) and one at $\nu = 1/7$ ($\mathrm{SU}(2)_5$). The predicted cross-correlation visibility is:

$$G_{2,3} = G_0 \left[1 + \mathcal{A}_2(j) \cdot \mathcal{A}_3(\ell) \cos(2\pi(e_2^* + e_3^*)V t/h)\right]$$

The product structure $\mathcal{A}_2 \cdot \mathcal{A}_3$ is the signature of adelic factorization. [speculative]

4.3 Distinguishing Adelic from Non-Adelic

A non-adelic anyon theory would produce $\mathcal{A}_{2,3} \neq \mathcal{A}_2 \cdot \mathcal{A}_3$ — correlation between the $p = 2$ and $p = 3$ anyon sectors that violates factorization. The ALP prediction is **exact factorization**:

$$\frac{\mathcal{A}_{2,3}(j,\ell)}{\mathcal{A}_2(j) \cdot \mathcal{A}_3(\ell)} = 1$$

A deviation from 1 would falsify the adelic framework. [my conjecture]

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5. Falsifiable Predictions

1. **[P1] Conductance visibility at $\nu = 1/4$.** The $p = 2$ anyon with $j = 1/2$ produces 88.8% visibility in FQHE interferometry at $\nu = 1/4$. [my conjecture]
   - **Falsification:** Measure visibility below 70% or above 95% at $\nu = 1/4$.
   - **[CHECK: 2028-06]**
2. **[P2] Factorization of place-crossing amplitudes.** $\mathcal{A}_{2,3}(j,\ell) = \mathcal{A}_2(j) \cdot \mathcal{A}_3(\ell)$ for all anyon types j, ℓ. [my conjecture]
   - **Falsification:** Measure $\mathcal{A}_{2,3}/(\mathcal{A}_2 \cdot \mathcal{A}_3) \neq 1$ at any (j,ℓ).
   - **[CHECK: 2029-12]**
3. **[P3] Hecke eigenvalue spectrum.** The Hecke eigenvalues $\alpha_p(j)$ (Phase 2.1) equal the interferometric phase shifts. [my conjecture]
   - **Falsification:** Compute $\alpha_p(j)$ from Verlinde algebra and find any discrepancy with measured phase shifts.
   - **[CHECK: 2028-12]**

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6. Conclusion

Place-crossing amplitudes provide the most direct experimental test of the Adelic Langlands Physics framework. The predicted 88.8% visibility at $\nu = 1/4$ ($p = 2$, $k = 2$) is within reach of current FQHE interferometry. The factorization $\mathcal{A}_{p,q} = \mathcal{A}_p \cdot \mathcal{A}_q$ is the signature of the adelic product structure — a deviation would falsify the framework. Phase 5 (Unification) synthesizes all predictions into a comprehensive experimental roadmap.

Declarations

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